
Canadian nutrient file:

database structure and file content description 2026



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Introduction

The Canadian Nutrient File (CNF) database relational files can be downloaded as Excel-compatible CSV delimited files available on the open government portal:

<https://open.canada.ca>

To view and use these files, you will need a database management software program or a spreadsheet program that can handle large files.

Database management programs

Some database management programs require you to "open" these files into their program while others require them to be "imported".

It is recommended to use the import feature to display the French accents properly.

Once opened or imported, the files can be joined using the common fields in graphic 1. Primary Key (PK) and Foreign Key (FK) are used to link support files.

Changes for this edition

The relational files content and its structure have changed significantly since the 2015 publication. For example:

- food codes are used as the primary key (PK) to link the main files (see graphic 1),
- alternate description and comment fields have been added in the food name file,
- yield amount, refuse amount and conversion factors have been merged into one measure weight conversion file,
- measure weight conversions are now published instead of conversion factors.

Due to changes in the file structure, files that track changes, additions, or deletions are not provided in this edition. We recommend downloading the entire database for reference. Field name correspondence from 2015 to 2026 edition and new added fields can be found in Appendix A.

Measure Weight Conversions

Portion size conversion factors from the 2015 publication are no longer used. They have been replaced by measure weight conversions where you do not have to multiply the portion size by 100 to know the weight of a measure.

For example, a user wants to know the weight of 250 ml of unsweetened applesauce (food code 1700):

- In the 2015 version, the conversion factor for that amount was equal to 2.57819.
- To obtain the weight you had to multiply this conversion factor by 100.
- In the CNF 2026, the measure weight conversion already includes the multiplication by 100 and is equal to 257.819 g.

Missing values

Not every food profile in the CNF contains data for all 173 nutrients. For example, one food profile might include information on 75 nutrients while another might have a complete set of data. Missing values mean data is unavailable, not that the nutrient content is zero.

Users need to understand and account for these missing values when using the database.

File Relationships

Graphic 1: CNF 2026 file relationships

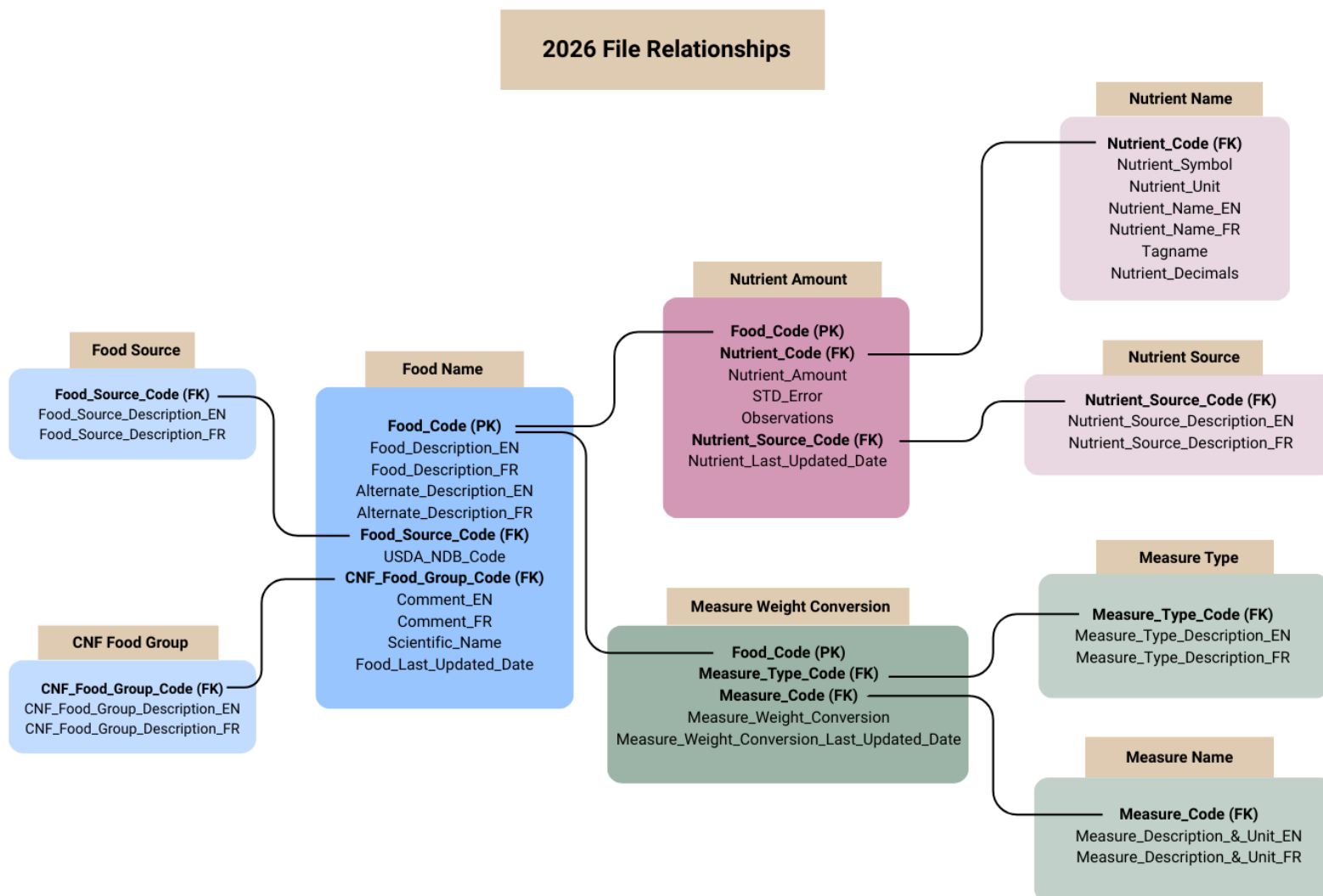


Table 1: General description of files

Name	File Name	File Description
Food Name	Food Name	This is a main file. This file contains all foods in the database. It includes the description of each food in both English and French, alternate descriptions in both languages, links to USDA data if present, comments, scientific name if present and the date the profile was last updated.
Food Source	Food Source	This file supports the Food Name file. It lists food sources in both English and French.
CNF Food Group	CNF Food Group	This file supports the Food Name file. It includes 23 food group headings in both English and French, based on similar food characteristics.
Nutrient Amount	Nutrient Amount	This is a main file. It links to the Food Name file (among others) to show which nutrients and amounts are recorded for each food.
Nutrient Name	Nutrient Name	This file supports the Nutrient Amount file. It includes abbreviations, nutrient names in both English and French, units of measure for each nutrient, Tagnames, and the number of decimal places.
Nutrient Source	Nutrient Source	This file supports the Nutrient Amount file. It lists the names of sources in both English and French.
Measure Weight Conversion	Measure Weight Conversions	This is a main file. It contains measure weight conversions and the date each conversion was last updated. Measure weight conversions include yield, refuse and commonly used measures. It links to the Food Name file.
Measure Type	Measure Type	This file supports the Measure Weight Conversion file. It contains a list of measure type descriptions in both English and French.
Measure Name	Measure Name	This file supports the Measure Weight Conversion file. It contains a list of measure descriptions and units, in both English and French.

Description of individual files

1. Food Name – Main File

N = Numeric field- # of characters / # of decimals (if decimal)

T = Text field

D = Date field

N/A = Not Applicable

File Name: Food Name

Field Name	Field Type	Field Size	Field Description	Relation to other files
Food_Code	N	Up to 4 digits integer	Four digit unique number. Primary Key and identifier code for the foods.	Nutrient Amount; Measure Weight Conversion
Food_Description_EN	T	Up to 512 characters	Main food name in English	N/A
Food_Description_FR	T	Up to 512 characters	Main food name in French	N/A
Alternate_Description_EN	T	Up to 512 characters	Additional information for the food in English. Can include key words, common names and/or brand names.	N/A
Alternate_Description_FR	T	Up to 512 characters	Additional information for the food in French. Can include key words, common names and/or brand names.	N/A
Food_Source_Code	N	Up to 2 digits integer	Foreign Key and identifier code for the food sources. Food sources indicate the origin, including Canadian content.	Food Source
USDA_NDB_Code	N	Up to 8 digits integer	Corresponds to the USDA NDB code.	N/A

Field Name	Field Type	Field Size	Field Description	Relation to other files
CNF_Food_Group_Code	N	Up to 2 digits integer	Foreign Key and identifier code for the Canadian Nutrient File food groups. There are 23 food groups.	CNF Food Group
Comment_EN	T	Up to 1024 characters	Additional information about the food in English.	N/A
Comment_FR	T	Up to 1024 characters	Additional information about the food in French.	N/A
Scientific_Name	T	Up to 255 characters	Scientific name of the food.	N/A
Food_Last_Updated_Date	D	yyyy-mm-dd	Date the food record was last updated in the database.	N/A

Additional information

Food Code

The food code is a four digit number which uniquely identifies each food, but doesn't describe or classify the food in any way. This unique code will not change over time.

Food Description

The "Food Name" file includes descriptive information about food items in English and French. Each description can be up to 512 characters and does not use abbreviations. A systematic hierarchy is used to record common food names, which may include details like product type, breed, part, physical state, shape or form, cooking method, preservation method, and brand name. For example:

- Chicken, broiler, thigh, meat and skin, stewed
- Cereal, ready to eat, Cheerios: honey nut, General Mills
- Soup, cream, mushroom, canned, condensed, reduced fat, 2% M.F. milk added

Alternate Food Description

New for this edition: Alternate food descriptions have been added to make it easier to search for foods using more natural language. These descriptions can include key words, common names or brand names. For example:

Food Code: 2920

Food Description: Carbonated drinks, cola, fast-food cola

Alternate Food Description: pop, soda, soft drink, fountain drink, Pepsi, Coke, Coca-Cola, Root Beer, Rootbeer, Dr Pepper, Doctor Pepper

USDA NDB Code

Previously known as “Country Code” in the 2015 publication, this code identifies whether a food comes from a USDA source and specifies which one. Note that not all CNF entries have a USDA NDB Code.

CNF Food Group Code

Foods are grouped into 23 categories based on similar characteristics.

1.1 Food Source - Support File to Food Name

File Name: Food Source

Field Name	Field Type	Field Size	Field Description	Relation to other files
Food_Source_Code	N	Up to 2 digits integer	Foreign Key and identifier code for the food sources. Food sources indicates the origin.	Food Name
Food_Source_Description_EN	T	Up to 255 characters	Food source description in English	N/A
Food_Source_Description_FR	T	Up to 255 characters	Food source description in French	N/A

1.2 CNF Food Group - Support File to Food Name

File Name: CNF Food Group

Field Name	Field Type	Field Size	Field Description	Relation to other files
CNF_Food_Group_Code	N	Up to 2 digits integer	Foreign Key and identifier code for the Canadian Nutrient File food groups. There are 23 food groups.	Food Name
CNF_Food_Group_Description_EN	T	Up to 255 characters	CNF food group name in English	N/A
CNF_Food_Group_Description_FR	T	Up to 255 characters	CNF food group name in French	N/A

2. Nutrient Amount – Main File

N = Numeric field- # of characters / # of decimals (if decimal)

T = Text field

D = Date field

N/A = Not Applicable

File Name: Nutrient Amount

Field Name	Field Type	Field Size	Field Description	Relation to other tables
Food_Code	N	Up to 4 digits integer	Four digit unique number. Primary Key and identifier code for the foods.	Food Name
Nutrient_Code	N	Up to 4 digits integer	Foreign Key and identifier code for nutrient name descriptions.	Nutrient Name
Nutrient_Amount	N	Up to 9 digits integer part, see Nutrient_Decimals for number of decimals in fractional part	Mean nutrient value in 100 g of edible portion of the food as described. The number of decimal places does not reflect the accuracy of the data.	N/A
STD_Error	N	Up to 4 digits integer part and 4 digits in fractional part	Standard error of the mean	N/A
Observations	N	Up to 4 digits integer	Number of samples	N/A
Nutrient_Source_Code	N	Up to 2 digits integer	Foreign Key and identifier code for nutrient source descriptions.	Nutrient Source
Nutrient_Last_Updated_Date	D	yyyy-mm-dd	Date the nutrient record was last updated in the database.	N/A

Additional information

Nutrient Amount

Not all foods in the CNF have values for every nutrient. If a nutrient value is missing, it doesn't mean the value is zero; it means no value is available for that nutrient.

Standard Error (STD Error)

Standard error calculated from the sample, sample composites, or contributing studies, provided the number of observations (n) is three or more.

Observations

The number of observations or samples the data is based on. If there are no standard error or number of samples, the values have been imputed or calculated from another form of the food, a similar food, or a calculated recipe.

2.1 Nutrient Name - Support File to Nutrient Amount

File Name: Nutrient Name

Field Name	Field Type	Field Size	Field Description	Relation to other files
Nutrient_Code	N	Up to 4 digits integer	Foreign Key and identifier code for nutrient name descriptions.	Nutrient Amount
Nutrient_Symbol	T	Up to 64 characters	Nutrient symbol or abbreviation of the nutrients. They may differ from international nomenclature.	N/A
Nutrient_Unit	T	Up to 256 characters	Unit of measure (e.g., mg, g, mcg or ug)	N/A
Nutrient_Name_EN	T	Up to 255 characters	Name of the nutrient in English.	N/A
Nutrient_Name_FR	T	Up to 255 characters	Name of the nutrient in French.	N/A
Tagname	T	Up to 64 characters	International Network of Food Data Systems (INFOODS) Tagnames [1]. A unique abbreviation for a food component developed by INFOODS to aid in the interchange of data.	N/A
Nutrient_Decimals	N	1 digit integer	Number of decimal places used in the rounding of the nutrient value.	N/A

2.2 Nutrient Source - Support File to Nutrient Amount

File Name: Nutrient Source

Field Name	Field Type	Field Size	Field Description	Relation to other files
Nutrient_Source_Code	N	Up to 2 digits integer	Foreign Key and identifier code for nutrient source descriptions.	Nutrient Amount
Nutrient_Source_Description_EN	T	Up to 255 characters	Nutrient source description in English.	N/A

Field Name	Field Type	Field Size	Field Description	Relation to other files
Nutrient_Source_Description_FR	T	Up to 255 characters	Nutrient source description in French.	N/A

Additional information

Nutrient Source Code

This code identifies the source of the nutrient amount data. The CNF provides a numeric code to each nutrient value, to indicate its source and type. The types of nutrient source codes can be one of the following:

Table 2 : Types of nutrient data sources

Type	Description
Analytical	Will show a mean, standard error and number of observations. Values are derived from chemical analysis of representative samples.
Imputed	Will only show a mean value. Assumptions have been made about the data by the compiler upon consultation with scientific experts or scientific literature.
Calculated	Will only show a mean value. No actual analyses are made but the calculations are straightforward. E.g. Soup diluted according to label specifications.
Assumed as zero	Will only show a value of zero. Assumptions have been made about the data by the compiler upon consultation with scientific experts or scientific literature.
Calculated from a recipe	Calculated value based on ingredient proportions. Retention factors and moisture losses are applied.
Provisional	May show a mean, standard error and number of observations, but there are questions surrounding the sampling and/or methods of analysis for these data, which remain to be verified.

3. Measure Weight Conversion – Main File

N = Numeric field- # of characters / # of decimals (if decimal)

T = Text field

D = Date field

N/A = Not Applicable

File Name: Measure Weight Conversion

Field Name	Field Type	Field Size	File Description	Relation to other tables
Food_Code	N	Up to 4 digits integer	Four digit unique number. Primary Key and identifier code for the foods.	Food Name
Measure_Type_Code	N	Up to 3 digits integer	Foreign Key and identifier code for type of measure for the measure weight conversion.	Measure Type
Measure_Code	N	Up to 6 digits integer	Foreign Key and identifier code for description of measure.	Measure Name
Measure_Weight_Conversion	N	Up to 9 digits integer part and 9 digits in fractional part	Weight of the measure	N/A
Measure_Weight_Conversion_Last_Updated_Date	D	yyyy-mm-dd	Date the measure weight conversion record was last updated in the database.	N/A

Additional information

Measure Weight Conversion

Nutrient values in the Canadian Nutrient File (CNF) are presented per 100 grams of the edible part of the food. For instance, the typical edible portions correspond to an apple without the core or a chicken leg without the bone. Actual weights may vary depending on the food source (e.g., organic or seasonal items). If exact values are important for your study, it is best to weigh the food yourself.

To find the nutrient value for a specific portion (e.g. household measure), use this formula:

$$N = V \cdot W / 100$$

Where:

N= nutrient value per portion (e.g. household measure)

V= nutrient value per 100 grams

W= measure weight conversion

For example, a Granny Smith apple, raw with skin (food code 7215), has 120 mg of potassium per 100 g. To find the amount of potassium in one medium Granny Smith apple which weighs 167 grams, the calculation is:

$$N = 120 \cdot 167 / 100$$

Where:

N= potassium value per one medium Granny Smith apple

120= potassium value per 100 grams

167= measure weight conversion of one medium Granny Smith apple

$$\text{Then } N = 200 \text{ mg}$$

All measurements are in the metric system. You can find metric conversion details in Appendix B. Lengths are given in millimetres (mm) or centimetres (cm).

3.1 Measure Type - Support File to Measure Weight Conversion

File Name: Measure Type

Field Name	Field Type	Field Size	File Description	Relation to other files
Measure_Type_Code	N	Up to 3 digits integer	Foreign Key and identifier code for type of measure for the measure weight conversion.	Measure Weight Conversion
Measure_Type_Description_EN	T	Up to 255 characters	Measure type description in English.	N/A
Measure_Type_Description_FR	T	Up to 255 characters	Measure type description in French.	N/A

Additional information

List of measure types

Refuse (Inedible Material)

Refuse refers to the parts of food that are not eaten, such as seeds, bones, and skin.

Here's how it is handled in the CNF:

- For raw meats, the items are raw as purchased;
- For cooked meats, the percent refuse is inedible material from the cooked state.
- For meat cuts containing bone, any connective tissue present is included in the value given for bone.
- Separable fat is not part of the refuse if the meat is described as "lean and fat". Lean refers to muscle tissue that can be readily separated out of the intact cut and includes any marbled fat within the muscle not removable by dissection.

User-defined

User-defined measures are those that can be helpful to the CNF users such as 250 ml or 1 unit of a fruit (e.g. 1 medium apple). They are often common measures used in various settings.

Yield

Sometimes, it is more useful to show the weight of the edible cooked food starting from a raw or raw-with-refuse state. These yields account for both:

- **Refuse losses** (like bones or peels), and
- **Cooking losses** (like moisture loss or evaporation).

In some cases, yields may also show **moisture gains**, such as when preparing foods from dry products (e.g., pudding mix).

Yields cannot be applied to data for the comparable raw food to “cook by calculation” as there are other factors such as nutrient retention to consider in such calculations.

3.2 Measure Name - Support Table to Measure Weight Conversion

File Name: Measure Name

Field Name	Field Type	Field Size	File Description	Relation to other files
Measure_Code	N	Up to 6 digits integer	Foreign Key and identifier code for description of measure.	Measure Weight Conversion
Measure_Description_&_Unit_EN	T	Up to 261 characters	Measure description and the number of units in that measure in English.	N/A
Measure_Description_&_Unit_FR	T	Up to 261 characters	Measure description and the number of units in that measure in French.	N/A

Appendix A: Field name correspondence between 2026 edition and 2015 edition

N/A = Not Applicable

File name: Food Name

CNF 2026 field name	Corresponding CNF 2015 field name
Food_Code	FoodCode
Food_Description_EN	FoodDescription
Food_Description_FR	FoodDescriptionF
Alternate_Description_EN (new)	N/A
Alternate_Description_FR (new)	N/A
Food_Source_Code	FoodSourceID
USDA_NDB_Code	CountryCode
CNF_Food_Group_Code	FoodGroupID
Comment_EN (new)	N/A
Comment_FR (new)	N/A
Scientific_Name	ScientificName
Food_Last_Updated_Date	FoodDateOfEntry

File name: Food Source

CNF 2026 field name	Corresponding CNF 2015 field name
Food_Source_Code	FoodSourceCode
Food_Source_Description_EN	FoodSourceDescription
Food_Source_Description_FR	FoodSourceDescriptionF

File name: CNF Food Group

CNF 2026 field name	Corresponding CNF 2015 field name
CNF_Food_Group_Code	FoodGroupCode
CNF_Food_Group_Description_EN	FoodGroupName
CNF_Food_Group_Description_FR	FoodGroupNameF

File name: Nutrient Amount

CNF 2026 field name	Corresponding CNF 2015 field name
Food_Code (new)	N/A
Nutrient_Code (new)	N/A
Nutrient_Amount	NutrientValue
STD_Error	StandardError
Observations	NumberOfObservation
Nutrient_Source_Code (new)	N/A
Nutrient_Last_Updated_Date (new)	N/A

File name: Nutrient Name

CNF 2026 field name	Corresponding CNF 2015 field name
Nutrient_Code	NutrientCode
Nutrient_Symbol	NutrientSymbol
Nutrient_Unit	Unit
Nutrient_Name_EN	NutrientName
Nutrient_Name_FR	NutrientNameF
Tagname	Tagname
Nutrient_Decimals	NutrientDecimals

File name: Nutrient Source

CNF 2026 field name	Corresponding CNF 2015 field name
Nutrient_Source_Code	NutrientSourceCode
Nutrient_Source_Description_EN	NutrientSourceDescription
Nutrient_Source_Description_FR	NutrientSourceDescriptionF

Appendix B: Metric system equivalents

Table 1: Metric system equivalents for units of measure

	Metric System Equivalents	Canadian Metric household measure		US and Imperial measures
Volume	4.9 ml	5 ml	n/a	1 teaspoon
	14.8 ml	15.0 ml	n/a	1 tablespoon
	29.57 ml	n/a	n/a	1 fluid ounce (US)
	28.41 ml	n/a	n/a	1 fluid ounce (Imperial)
	236.6 ml	250 ml	n/a	1 cup (8 US fluid ounces)
	473.2 ml	n/a	n/a	1 pint (16 US fluid ounces)
	568.3 ml	500 ml	n/a	1 pint (20 Imperial fluid ounces)
	946.4 ml	n/a	n/a	1 quart (32 US fluid ounces)
	1136.5 ml	1 L	n/a	1 quart (40 Imperial fluid ounces)
	3786 ml	n/a	n/a	1 gallon (128 US fluid ounces)
	4546 ml	4 L	n/a	1 gallon (160 Imperial fluid ounces)
	16.39 ml	2.54 cm cube	15.63 ml	1 cubic inch
Length	2.54 cm, 25.40 mm	n/a	n/a	1 inch
Weight	28.35 g	n/a	n/a	1 ounce
	453.6 g	n/a	n/a	1 pound
	140 g	250 ml	148 g	1 cup (poultry and cooked meats chopped and diced)
	110 g	250 ml	116 g	1 cup (poultry and cooked meats ground)
Energy	4.184 kJ	n/a	n/a	1 Calorie (1 kcal)

Table 2: List of Definitions

	Definition
Heaping teaspoon	Refers to an ordinary teaspoon rather than to a standard measuring teaspoon
Not packed	Lightly filled measure without pressing down on the food
Packed	Maximum amount of food that can be pressed into the measure without altering its physical structure
Pared	Skin removed plus some adhering flesh
Peeled	Skin removed with a minimum of adhering flesh

References

1. FAO/INFOODS. 2012. Tagnames for Food Components. Version current 2014.
<http://www.fao.org/infoods/infoods/standards-guidelines/food-component-identifiers-tagnames/en/>

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